IMPaCT-Data Biomedical Cloud

Towards a federated virtual computing multi-environment for Precision Medicine in Spain.

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I. BACKGROUND

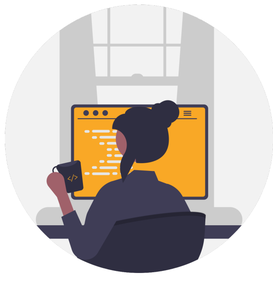
The Spanish Precision Medicine Infrastructure associated with Science and Technology (**IMPaCT**), aims to impulse **precision medicine** within the Spanish National Health System. **IMPaCT-Data**, the Data Science pillar of this infrastructure, has been tasked to design, deploy and extend a **Biomedical Cloud** taking as reference similar efforts across Europe.

II. OBJECTIVE

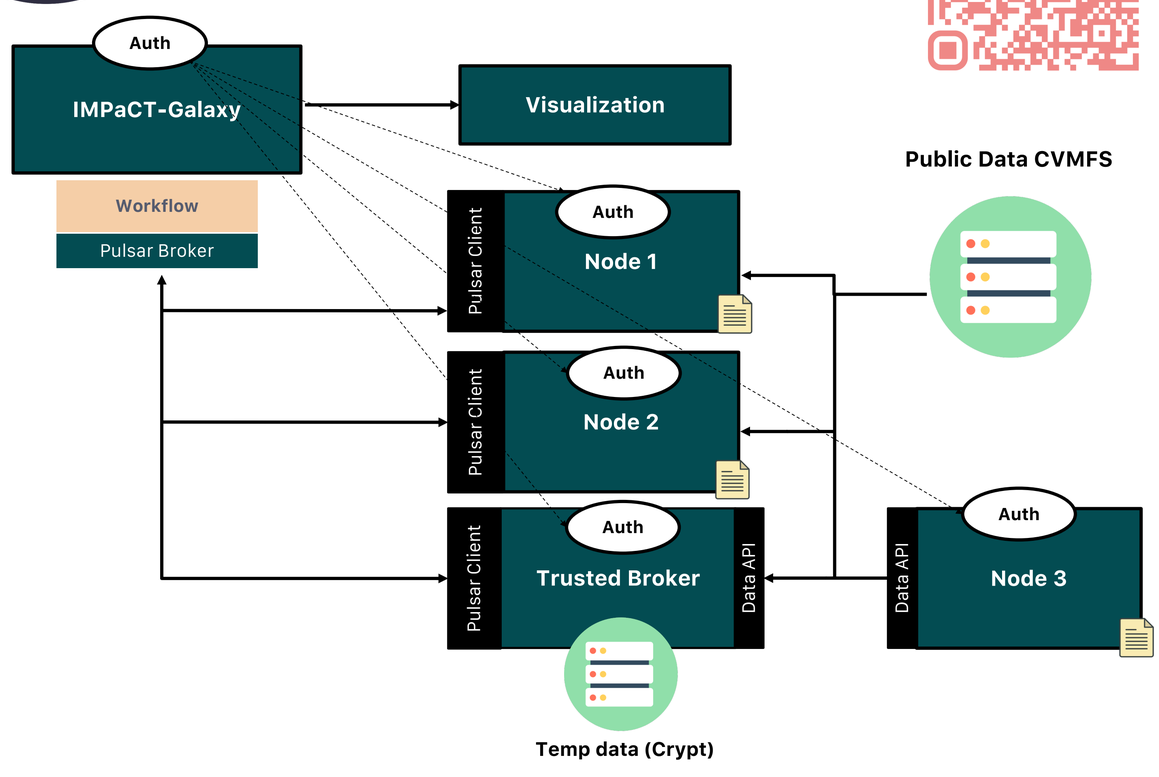
The establishment of IMPaCT-Data **Biomedical Cloud** aims to provide a **scalable** and **flexible** analysis **environment**, enabling the **integration**, **management** and **analysis** of sensitive and non-sensitive data, e.g. richly described genomics, structured clinical, and medical imaging data, made available within IMPaCT for research purposes.

III. ARCHITECTURE

The infrastructure is designed as a **federated cloud system,** which includes a *Life Sciences Login*-compliant **authentication system**, a data infrastructure to manage both **public and sensitive data**, and a **distributed** set of **computational resources**, which will be assigned depending on the data that should be mobilized, and the needs of the desired analysis.

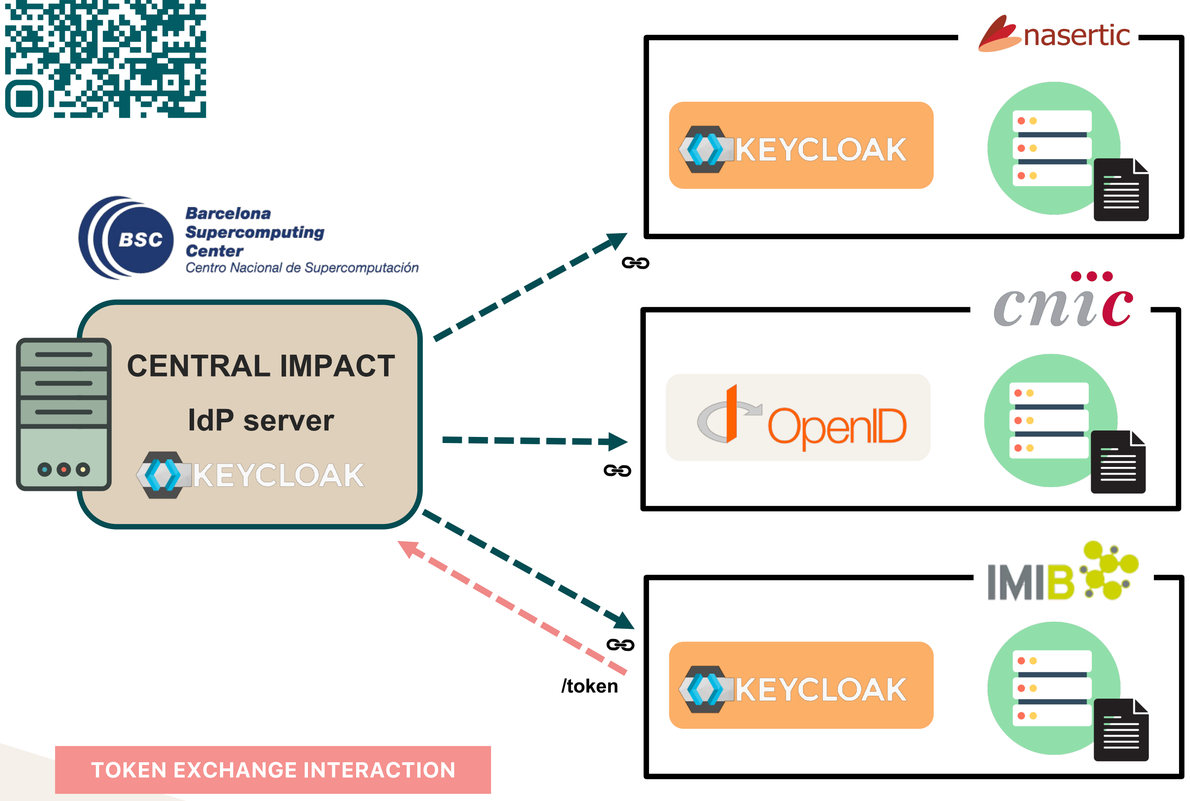


INFRASTRUCTURE CONCEPTION



FEDERATED AUTHORIZATION SYSTEM

THE NEW KID IN THE BLOCK



TOKEN EXCHANGE INTERACTION

Each node maintains its **own authentication server** linked to the central federated server (IMPaCT).

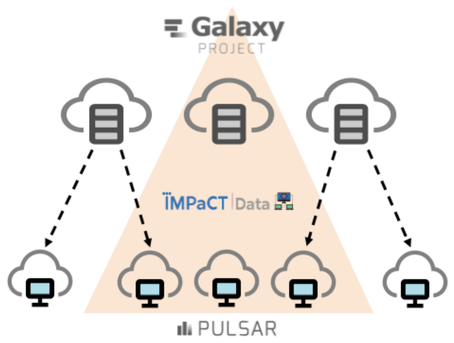
Authorization assertions are provided by each node in the form of **GA4GH Passport v1 OIDC** claims.

OIDC Token Exchange is used to exchange IMPaCT tokens for tokens **issued by specific nodes**.

CURRENT PLATFORMS

GALAXY

- Structure based on the connection of **one main node** (IMPaCT) and **three pulsar workers** (BSC, IMIB, CNIC)
- **Parallel jobs** proceeding from the execution of a **single tool** within the IMPaCT environment can be executed in **different cores.**
- Pulsar nodes can be connected to **several main** Galaxy servers, even if they are outside of the IMPaCT network and receive jobs from both of them **simultaneously**.



VRE

One of the latest platforms being incorporated to the IMPaCT-Data Biomedical Cloud is a dedicated instance of the **Virtual Research Environment (VRE).** Based on openVRE, it provides a **personal workspace** to combine tools and data and to develop **distributed analysis workflows** in a secure manner.

…AND MORE TO COME! As part of some of other developments taking place right now a **reference implementation** of all the recipes and configuration is also being developed by the IMPaCT-Data team to ensure enviromnent **reproductibility**.

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The project IMPaCT-Data (Exp. IMP/00019) has been funded by the Institute of Health Carlos III, co-funded by the European Union, European Regional Development Fund (ERDF, "A way to make Europe").